School Worksheet

Class 12 Mathematics, Chapter 11: Three Dimensional Geometry

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The direction cosines of a line equally inclined to the coordinate axes are:

(A) $\pm 1, \pm 1, \pm 1$

(B) $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

(C) $\pm\frac{1}{3}, \pm\frac{1}{3}, \pm\frac{1}{3}$

(D) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}$

2 MCQ — 1 mark

If a line makes angles 90°, 135° and 45° with the x, y and z axes respectively, its direction cosines are:

(A) $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(B) $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(C) $1, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(D) $0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$

3 MCQ — 1 mark

The Cartesian equation of the line through $(5, -2, 4)$ and parallel to $3\hat{i} + 2\hat{j} - 8\hat{k}$ is:

(A) $\frac{x+5}{3} = \frac{y-2}{2} = \frac{z+4}{-8}$

(B) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{-8}$

(C) $\frac{x-3}{5} = \frac{y-2}{-2} = \frac{z+8}{4}$

(D) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{8}$

4 MCQ — 1 mark

The angle between the lines $\frac{x-2}{2} = \frac{y-1}{5} = \frac{z+3}{-3}$ and $\frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$ is:

(A) $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$

(B) $\cos^{-1}\left(\frac{26}{9}\right)$

(C) $\cos^{-1}\left(\frac{13}{9\sqrt{38}}\right)$

(D) $\frac{\pi}{2}$

Q No.	Question	Marks
5	MCQ The value of p for which the lines $\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ are perpendicular is: **(A)** $\frac{70}{11}$ **(B)** $\frac{11}{70}$ **(C)** $-\frac{70}{11}$ **(D)** 7	1 mark
6	Assertion-Reason **Assertion (A):** The line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ passes through the origin. **Reason (R):** Any line of the form $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ passes through the origin, since $(0,0,0)$ satisfies it. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find the vector and Cartesian equations of the line passing through $A(1,2,-1)$ and $B(2,1,1)$.	2 marks

Level 2

16 marks

8	Short answer Show that the lines $\frac{x-5}{7}=\frac{y+2}{-5}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ are perpendicular to each other.	2 marks
9	Short answer Find the shortest distance between the lines $\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k})$.	3 marks
10	Short answer Find the distance between the parallel lines $\vec{r}=\hat{i}+2\hat{j}-4\hat{k}+\lambda(2\hat{i}+3\hat{j}+6\hat{k})$ and $\vec{r}=3\hat{i}+3\hat{j}-5\hat{k}+\mu(2\hat{i}+3\hat{j}+6\hat{k})$.	3 marks
11	Short answer Find the angle between the lines $\vec{r}=3\hat{i}+2\hat{j}-4\hat{k}+\lambda(\hat{i}+2\hat{j}+2\hat{k})$ and $\vec{r}=5\hat{i}-2\hat{j}+\mu(3\hat{i}+2\hat{j}+6\hat{k})$.	2 marks
12	Short answer Find k so that the lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-5}$ are perpendicular.	2 marks
13	Short answer Find the direction cosines of the line whose Cartesian equations are $5x-3=15y+7=3-10z$.	2 marks

Q No.	Question	Marks
14	Short answer Find the Cartesian equation of the line through $(-2, 4, -5)$ and parallel to $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$.	2 marks

Level 3

27 marks

15	Long answer Find the coordinates of the foot of the perpendicular drawn from $P(1, 2, 3)$ to the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Also find the length of the perpendicular.	5 marks
16	Long answer Find the image A' of the point $A(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Also, find the equation of the line joining A and A'.	5 marks
17	Long answer Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect. Find their point of intersection.	5 marks
18	Long answer Find a point P on the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ such that its distance from the point $Q(2, 4, -1)$ is 7 units. Also, find the equation of the line joining P and Q.	5 marks
19	Short answer Find the vector equation of the line passing through $(1, 2, -4)$ and perpendicular to both the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.	3 marks
20	Case study **Case Study:** Two aircraft fly along straight paths given by $L_1 : \vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $L_2 : \vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$. Air-traffic control needs to know how close the paths come.	4 marks

(i) Write the direction ratios of L_1 and L_2. — 1 mark

(ii) Are the paths parallel? Justify. — 1 mark

(iii) Find the shortest distance between the two paths. — 2 marks

Answer key

Chapter 11: Three Dimensional Geometry, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$
2	(B) $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$
3	(B) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{-8}$
4	(A) $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$
5	(A) $\frac{70}{11}$
6	(A) Both A and R are true and R is the correct explanation of A.
7	$\vec{r} = (\hat{i}+2\hat{j}-\hat{k}) + \lambda(\hat{i}-\hat{j}+2\hat{k})$; $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z+1}{2}$

Level 2

8	$7(1)+(-5)(2)+1(3) = 0$, so the lines are perpendicular.
9	Shortest distance $= \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$ units
10	Distance $= \frac{\sqrt{293}}{7}$ units
11	$\theta = \cos^{-1}\left(\frac{19}{21}\right)$
12	$k = -\frac{10}{7}$
13	$\frac{6}{7}, \frac{2}{7}, -\frac{3}{7}$
14	$\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$

Level 3

15	Foot of perpendicular $M(3,5,9)$; length of perpendicular $= 7$ units
16	Image $A'(1,0,7)$; line AA': $\vec{r} = (\hat{i}+6\hat{j}+3\hat{k}) + \mu(-3\hat{j}+2\hat{k})$, i.e. $\frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$
17	The lines intersect at $\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$.
18	$P(-4,1,-3)$; line PQ: $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$
19	$\vec{r} = (\hat{i}+2\hat{j}-4\hat{k}) + \lambda(2\hat{i}+3\hat{j}+6\hat{k})$
20	(i) $2, -1, 1$ and $3, -5, 2$; (ii) No, DRs are not proportional; (iii) $\frac{10}{\sqrt{59}}$ units